

RESEARCH INTERESTS	Human–Robot Interaction*, Soft Robotics, Robot Control and Learning, Autonomy for Mobility and Manipulation, Healthcare Robotics, AI Ethics, and Industrial Robotics.	
EDUCATION	Worcester Polytechnic Institute, Massachusetts, United States Master's, Robotics Engineering 4.0/4.0 CGPA	Jan 2026 – Dec 2026
	Pan-Atlantic University, Lagos, Nigeria - Bachelor of Engineering in Mechanical Engineering - First Class, 4.79/5.0 CGPA	Jan 2021 – July 2025
PUBLICATION & PREPRINTS	Samuel Oyefusi, Raehan Kareem, Onoriode Avebenake, "Application of Vector Integral Theorems in Engineering" Vol. 1 No. 2: 2026 - Alexander Ozoani, Samuel Oyefusi "Mortality of Distributed Understanding" Human Arenas, 1-7, Springer Nature Samuel Oyefusi, Abima Cynthia Samuel, "Microstructure Property Relationships in Industrial Chromium Containing Ferrous Alloys Under Annealed and Hardened States," In review Journal of Materials Engineering and Performance. - Chima Okwuokei , Desmond Moru , Clifford Uroh, Samuel Oyefusi, "Virtual Reality Training System for Enhancing Skill Transferability in Volleyball Players ", In review Journal of Virtual worlds, 2026, MDPI. Samuel. Oyefusi, Darlington Agholor, Lawrence. Ukpoju, Peter Onu "Development of a Cost-Effective Autonomous Ultraviolet-C Disinfection and Vacuum-Cleaning Robot," ICARA, 2026 Samuel Oyefusi, Oluwarantimi Bukola, Peter Onu, "The Potential of Generative Artificial Intelligence to Improve Learning in Schools" The 3rd International Conference on Sustainability: DEVELOPMENTS & INNOVATIONS, Springer Nature, (Under publishing). - Samuel Obiagba, Samuel Oyefusi, Peter Onu, "Easy Inertial Measurement Unit (EIMU): A Cost Effective IMU Solution for Real-Time Orientation and Acceleration Estimation in Robotic State Estimation and ROS2 System " The 8th International Conference on innovative and Smart Materials (Under publishing). - Samuel Obiagba, Samuel Oyefusi, Peter Onu, "Easy PID Motor Controller (EPMC): A Cost Effective Closed-Loop PID Motor Controller with Integrated Encoder Processing and ROS2 Support for Mobile Robotics" The 8th International Conference on innovative and Smart Materials (Under publishing). Samuel Oyefusi, Peter Onu, "Value Before Autonomy: A Human-Centered Model for Consumer Robots" In review International Journal of Social Robotics, Springer Nature - Tito Osadebey, Samuel Oyefusi, Micah Udeogu, "Addressing Artificial Intelligence Bias through Inclusivity: A Case Study with Nigerian Food Images," 2025 International Conference on Artificial Intelligence and Digital Ethics (ICAIDE), Guangzhou, China, 2025, pp. 469-473, doi: 10.1109/ICAIDE65466.2025.11189650. - Peter. Onu, Anup. Pradhan, Nelson. Madonsela and Samuel. Oyefusi, "Designing Resilient AI-Powered Digital Assistants for Industrial Applications: A Mini Review of Key Challenges," IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), 2025.	

PROFESSIONAL SERVICES

Reviewer:

- International Journal of Social Robotics, Springer Nature 2026-present
- International Conference on Industry of the Future and Smart Manufacturing 2025- present
- The Artificial Life Conference, University of Waterloo 2026- present
- ROSCon Diversity Scholarship Committee , Open Robotics 2026- present
- Journal of Computer Science, Science Publications 2025-Present
- Journal of Mechatronics and Robotics, Science Publications 2025-Present

PROFESSIONAL EXPERIENCE

Project Engineering Intern – TechnipFMC Aug. 2025 – Dec. 2025

- Assisted Project Engineers in developing project plans, schedules, and resource allocations.
- Supported creation of technical documentation including drawings, specifications, and progress reports.

Intern, Pipeline & Gas Supply – Nigeria LNG Limited | Nigeria Jul. 2024 – Sept. 2024

- Automated manual reports and used advanced data analytics tools to analyse 25 years of pigging operations data, uncovering trends in pipeline wear, cleaning efficiency, and corrosion patterns, resulting in improved predictive maintenance planning and reduced downtime risks.
- Developed an automated data logging system to eliminate hard-to-trace data, improving data accuracy, accessibility, and real-time monitoring of pipeline conditions.

Robotics Engineering Intern – Orangewood | Remote May 2024 – Jul. 2024

- Developed Advanced Manipulation Applications to solve real-world problems, enhancing robotic dexterity and efficiency in industrial environments.
- Contributed to the development of Robo-GPT, refining its functionality to streamline automation and improve user experience.
- Collaborated with AI and engineering teams to develop robotic simulators, enhancing real-world testing environments for cobot performance evaluation.
- Integrated MoveIt, ROS, and Gazebo simulation to improve robot manipulator performance, enabling more efficient testing and validation.

AWARDS

- Most Innovative Tech Award Male – Pan-Atlantic University 2025
- Leadership Award Nominee – Pan-Atlantic University 2025
- 2nd Overall best Student, Faculty of Engineering – Pan-Atlantic University 2025
- 2nd best Student Paper – First International Conference on Governance, Ethics and Technology 2025
- Mr. SST Award – Pan-Atlantic University 2025
- Awarded for exemplifying the School of Science and Technology’s spirit through leadership, integrity, and community impact
- 2nd Place African Mars Rover Competition– African Rover Challenge 2024
- Innovator of the Year– Pan Atlantic University 2024
- Young Innovators in Africa Nominee – African Innovation Conference 2023
- Pan African Robotics Competition Finalist – PARC 2024

PROFESSIONAL MEMBERSHIPS

- American Association for the Advancement of Science (AAAS)
- National Society of Professional Engineers
- Project Management Institute
- Black in AI
- National Postdoctoral Association
- National Society of Black Engineers
- Black in Robotics

TECHNICAL SKILLS	• Programming languages: Python, C++, MATLAB, R • Design: Fusion360, SolidWorks, AutoCAD, Inventor • Frameworks: ROS2, PyTorch, NumPy, matplotlib, Pandas, OpenCV, Yolo • Hardware Design: Simscape, NI Multi-Sim • Simulation: Ansys, Gazebo, Omniverse, Isaac-sim • Version Control: Git, Bitbucket	
MEDIA FEATURE	#ResearchDiary2: How Samuel Oyefusi is Building Robots Against the Odds Neural Narratives "Meet Samuel Oyefusi, innovator providing affordable healthcare solutions," BusinessDay "PAU students develop the first in Africa prototype of Mars rover" – BusinessDay, Channels "Stem From, Samuel Oyefusi" – YouTube, Spotify, "Africa Artificial Intelligence" – <i>Youth and Entrepreneurs Embracing Technology</i> – Al Jazeera,	
SPEAKING ENGAGEMENT	Presenter – "Technology" – First International Conference on Governance, Ethics and Technology (GET1), Speaker – "Empowering Children and Young People to Lead Climate Action" UNICEF World Children's Day. Panellists – "Fostering Youth-led Innovation for Agrifood System Transformation" United Nations World Food Forum. Speaker – "The Future of Africa's Space Industry: Capacity Building for Sustainable Development" – SGAC Speaker – "Transition, Mastery & Momentum in Engineering" – Lagos State University.	
TEACHING	Teaching Assistant, Elevate Them • GET 501 Introduction to Applied Design • GET 503 Introduction to Engineering Simulation	2022-2025
	Summer Course Instructor, Black in robotics, Introduction to ROS2,	2025
	Undergraduate Teaching Assistant, Pan-Atlantic University • MEE 57 Industrial Automation and Control • MCT 401 Introduction to Mechatronics • MEE 501 Design of Robotics and Control Systems Peer Tutoring Program, Pan-Atlantic University • Provided academic support to junior and peer students in core Mechanical Engineering courses through structured tutorials and mentoring sessions.	2022-2025
OTHER ACTIVITIES	Member, Young People Action Team – Generation Unlimited Nigeria • Developed and distributed skill-building resources to empower youth, enhancing their employability and access to opportunities.	Jan. 2025 – Jan. 2026
	Alumni Class President – Pan Atlantic University Nigeria • Organised industry excursions and hosted conferences with industry leaders, bridging the gap between academia and the professional world. • Led fundraising efforts for Final Year Week, securing resources for memorable celebrations and networking events. • Represented PAU at alums and decision-making events, advocating for student interests and strengthening alum relations.	Nov. 2024 – Present
	President – Living Green Club Nigeria • Led sustainability initiatives, implementing innovative irrigation systems, flood prevention, and an AI-powered Agrobot to drive eco-friendly solutions. • Managed CSR collaborations, engaging corporate partners and organising workshops to promote climate action and sustainability on campus. • Coordinated and motivated club members and stakeholders in environmental awareness and advocacy.	Jul. 2021 – Sept. 2024